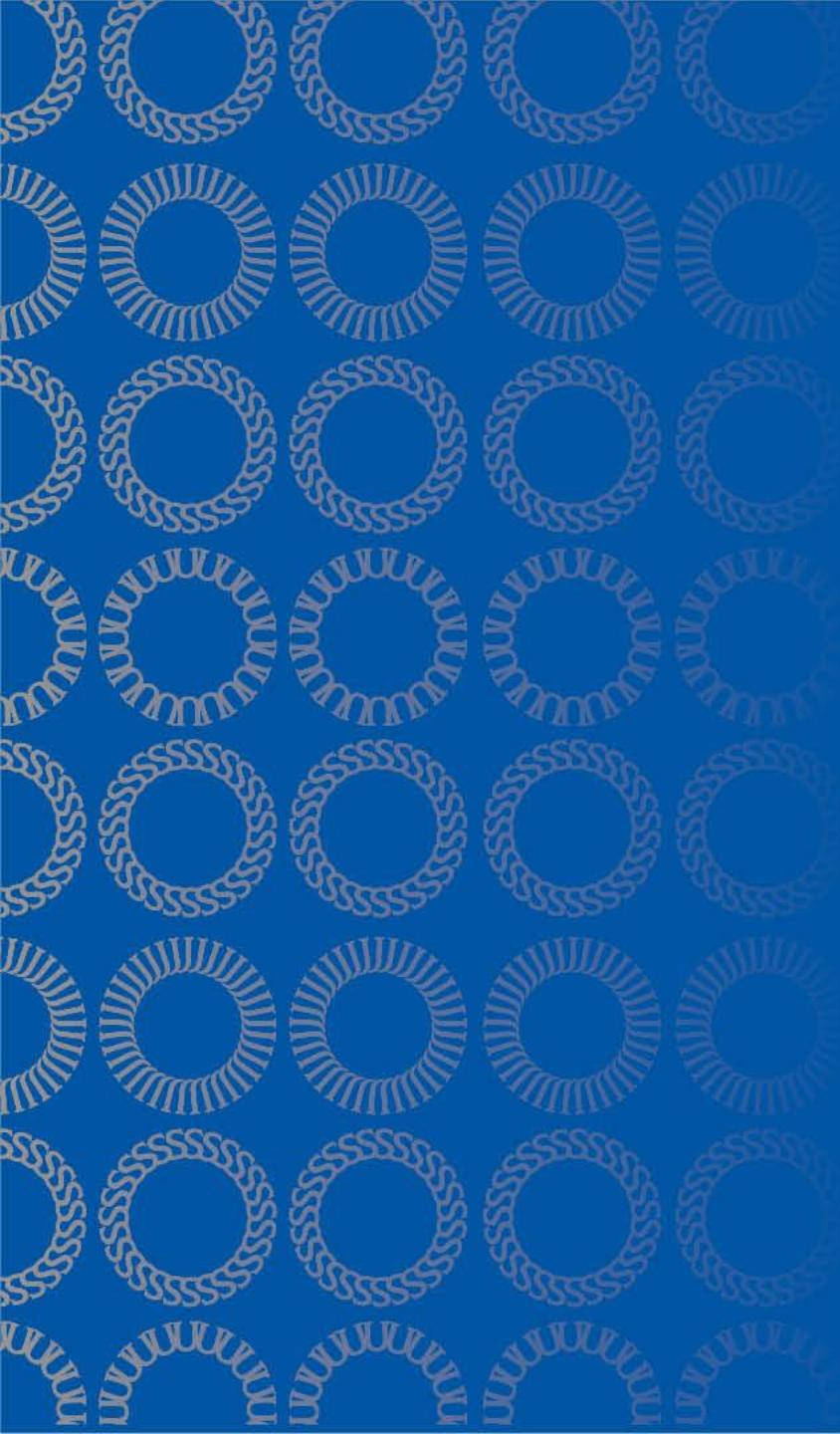




DATA 255 Deep Learning Technologies

Sep 25, 2025

Taehee Jeong, Ph.D.

Today's Topics(1)

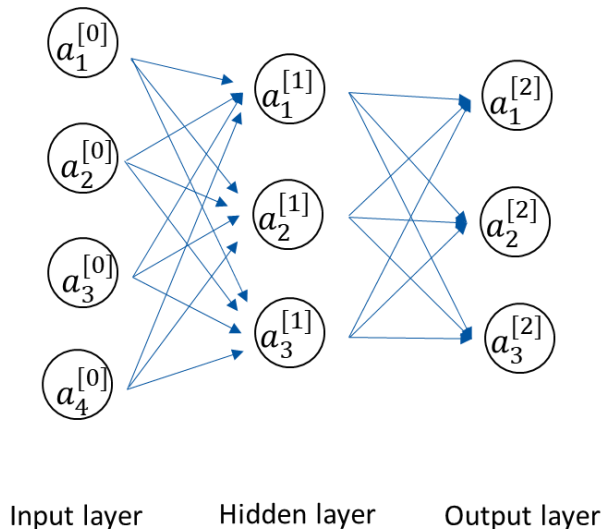
- Deep neural network (DNN) vs. Convolutional neural network (CNN)
- Convolution layer

Deep neural network (DNN)

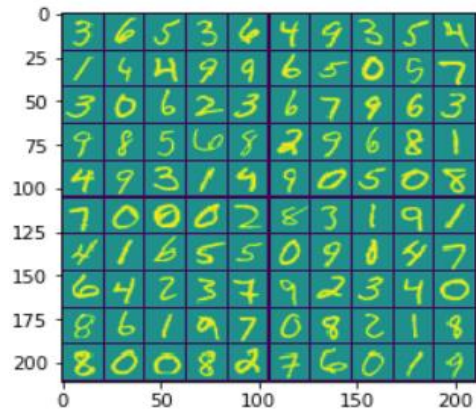
; Artificial neural network (ANN)

It can be applied to regression or classification.

- Regression: predict continuous valued output
- Classification: predict categorical value



Digital representation of an image



	0	1	2	3	4	5	6	7	8	9	...	391	392	393	394	395	396	397	398	399	y
0	0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	5
1	0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	9
2	0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	7
3	0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	6
4	0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	5

- Grayscale image is a matrix of pixels (**picture elements**)
- Dimensions of this matrix are called image resolution (e.g. 28 x 28)
- Each pixel stores its brightness (or **intensity**) ranging from 0 to 255, 0 intensity corresponds to black color
- Color images store pixel intensities for 3 channels: red, green and blue

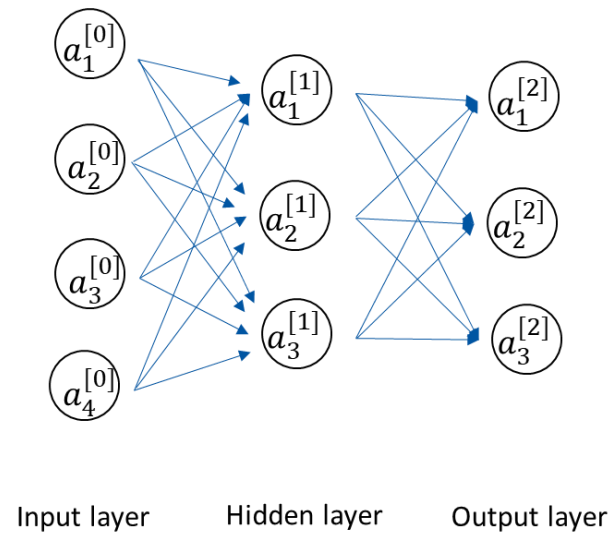
Why not Deep Neural Network?

- Why not simply use a regular deep neural network with fully connected layers for image recognition tasks?
- Unfortunately, although this works fine for small images (e.g., MNIST), it breaks down for larger images because of the huge number of parameters it requires.
- For example, a 100×100 image has 10,000 pixels, and if the first layer has just 1,000 neurons (which already severely restricts the amount of information transmitted to the next layer),
- this means a total of 10 million connections. And that's just the first layer.
- CNNs solve this problem using [partially connected layers](#).

Why not Deep Neural Network?



$1000 \times 1000 \times 3 = 3\text{M}$



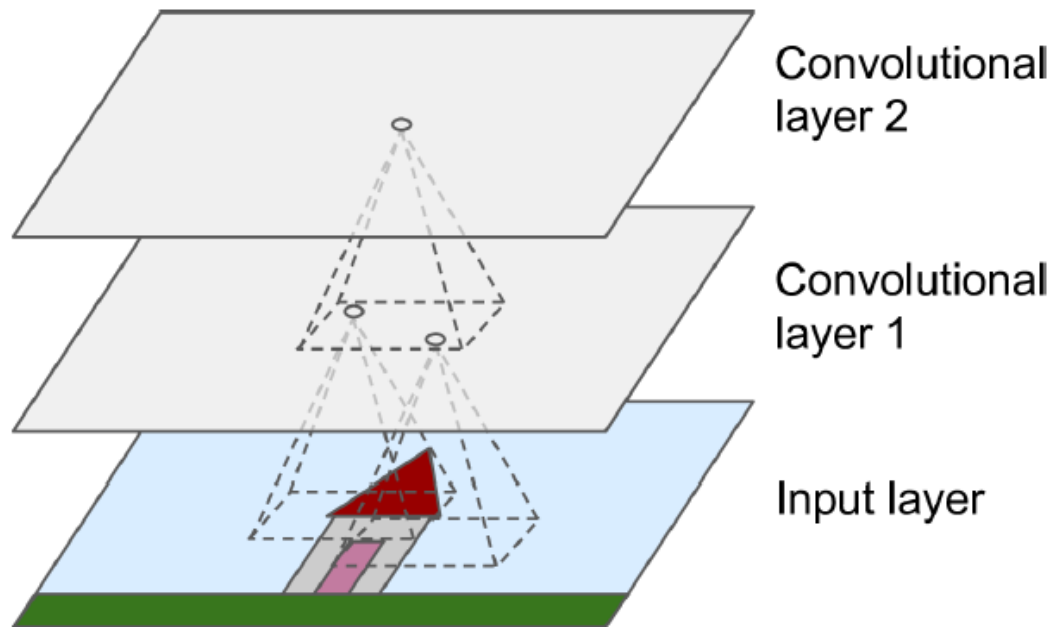
Convolutional Layer

- The most important building block of a CNN is the *convolutional layer*:
- Neurons in the first convolutional layer are not connected to every single pixel in the input image, but only to pixels in their receptive fields.
- Neuron in the second convolutional layer is connected only to neurons located within a small rectangle in the first layer.
- This architecture allows the network to concentrate on low-level features in the first hidden layer, then assemble them into higher-level features in the next hidden layer, and so on.
- This hierarchical structure is common in real-world images, which is one of the reasons why CNNs work so well for image recognition.

Advantages of CNN over DNN for image classification

- CNN has **fewer parameters** than a fully connected DNN
- CNN can detect a particular **feature anywhere** on the image instead of detecting only in that **particular location**.
- CNN's architecture is able to identify pixels' organization (how each pixel is located).

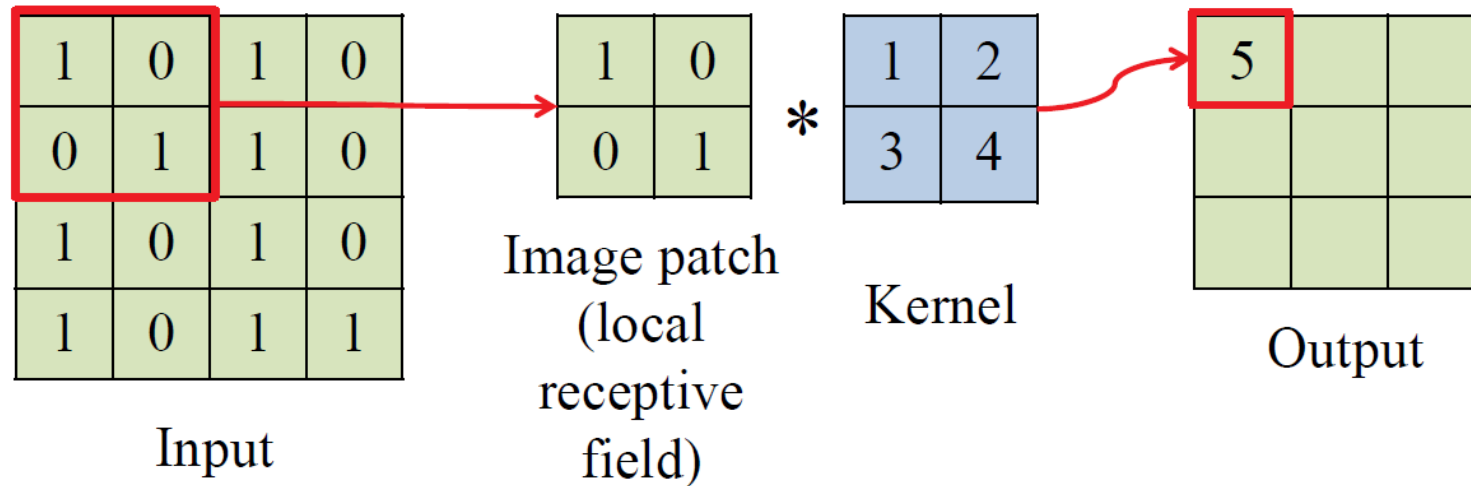
Convolutional layer



Source: *Hands-On Machine Learning with Scikit-Learn & TensorFlow*

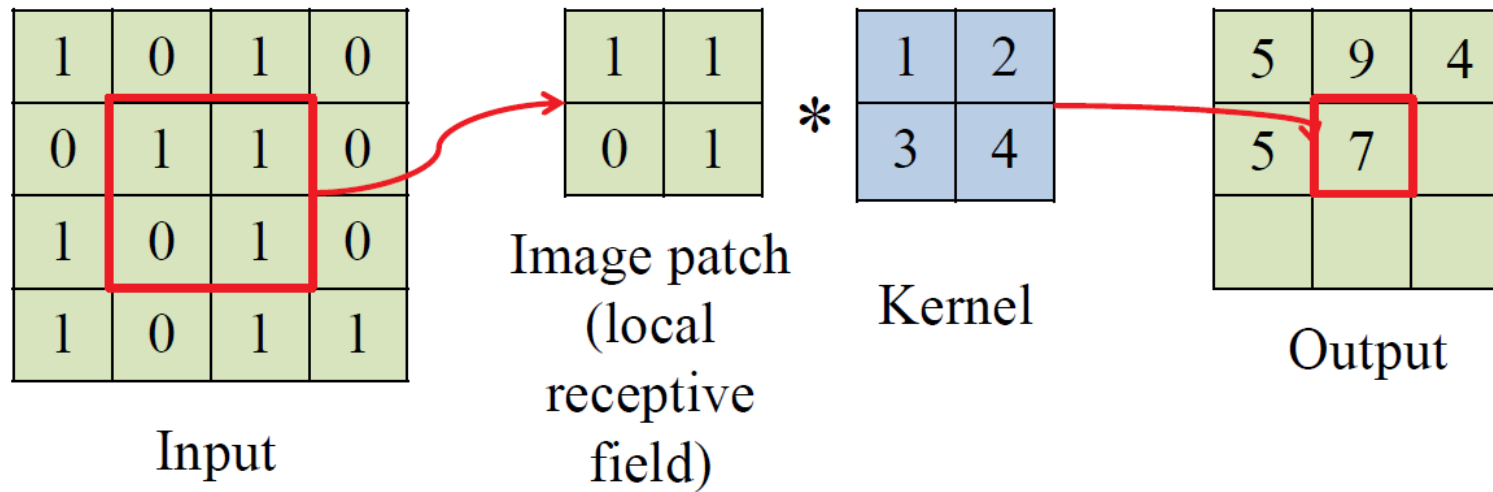
Convolutions

Convolution is a dot product of a **kernel** (or filter) and a patch of an image (**local receptive field**) of the same size



Convolutions

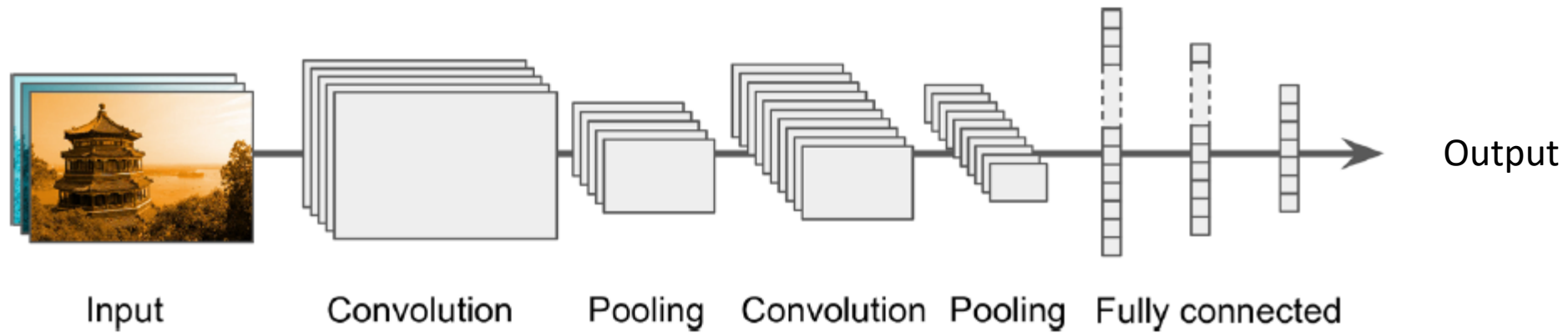
Convolution is a dot product of a **kernel** (or filter) and a patch of an image (**local receptive field**) of the same size



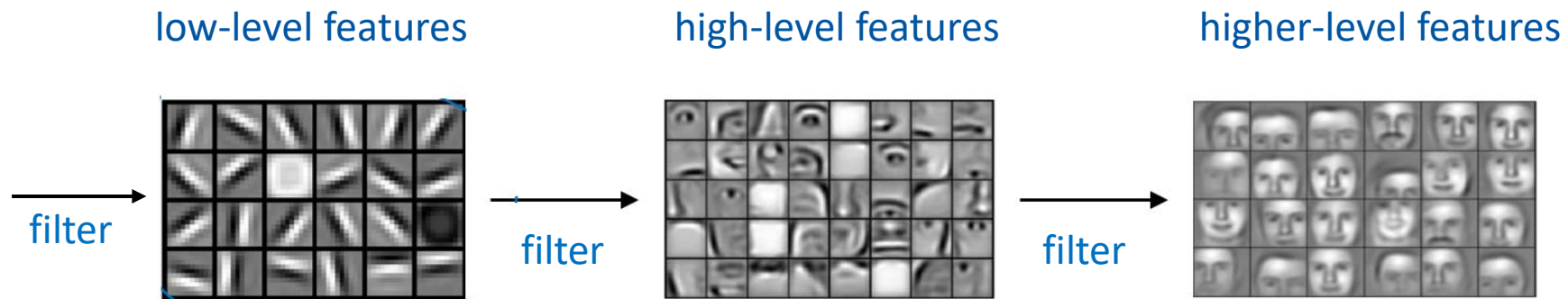
Today's Topics(2)

- Operations in CNN
 - Convolution (CONV)
 - Padding
 - Stride
 - Pooling (POOL)
 - Fully connected (FC)

CNN Architecture

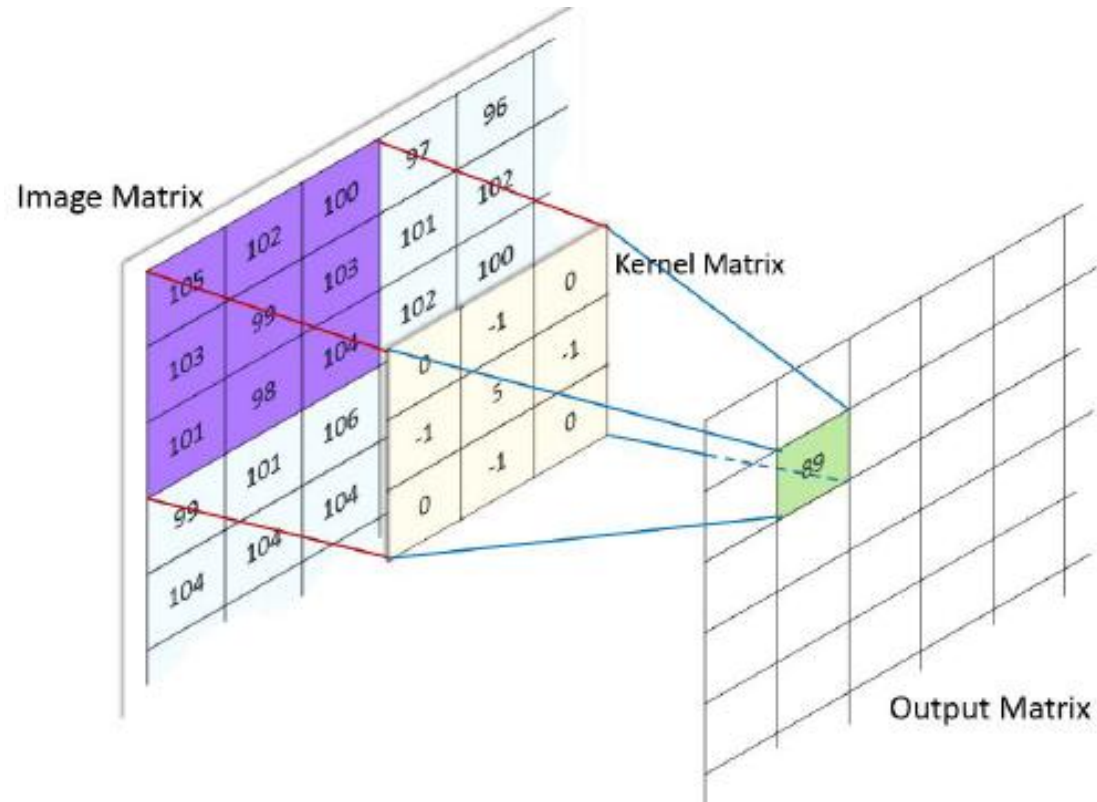


Convolutional layer



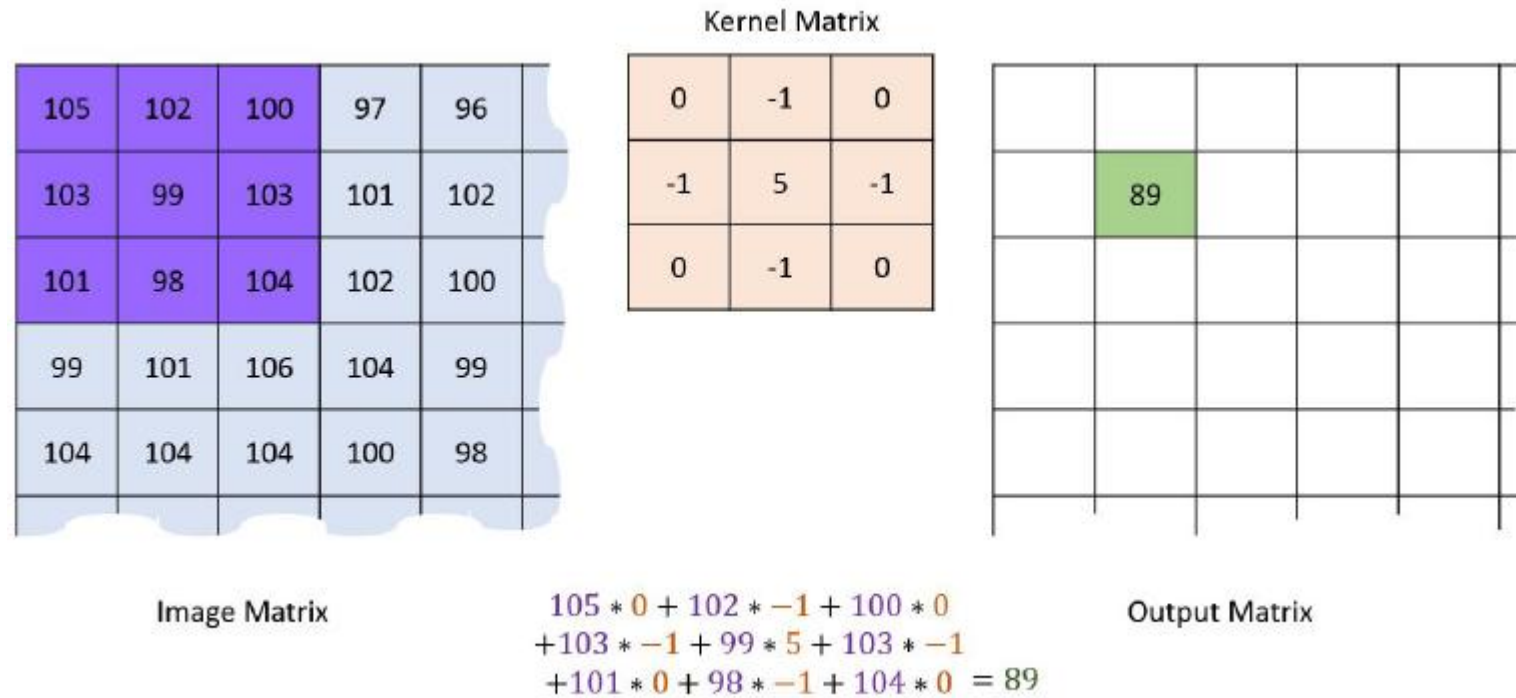
Source: Deep Neural Network, Andrew Ng

Image convolution operation



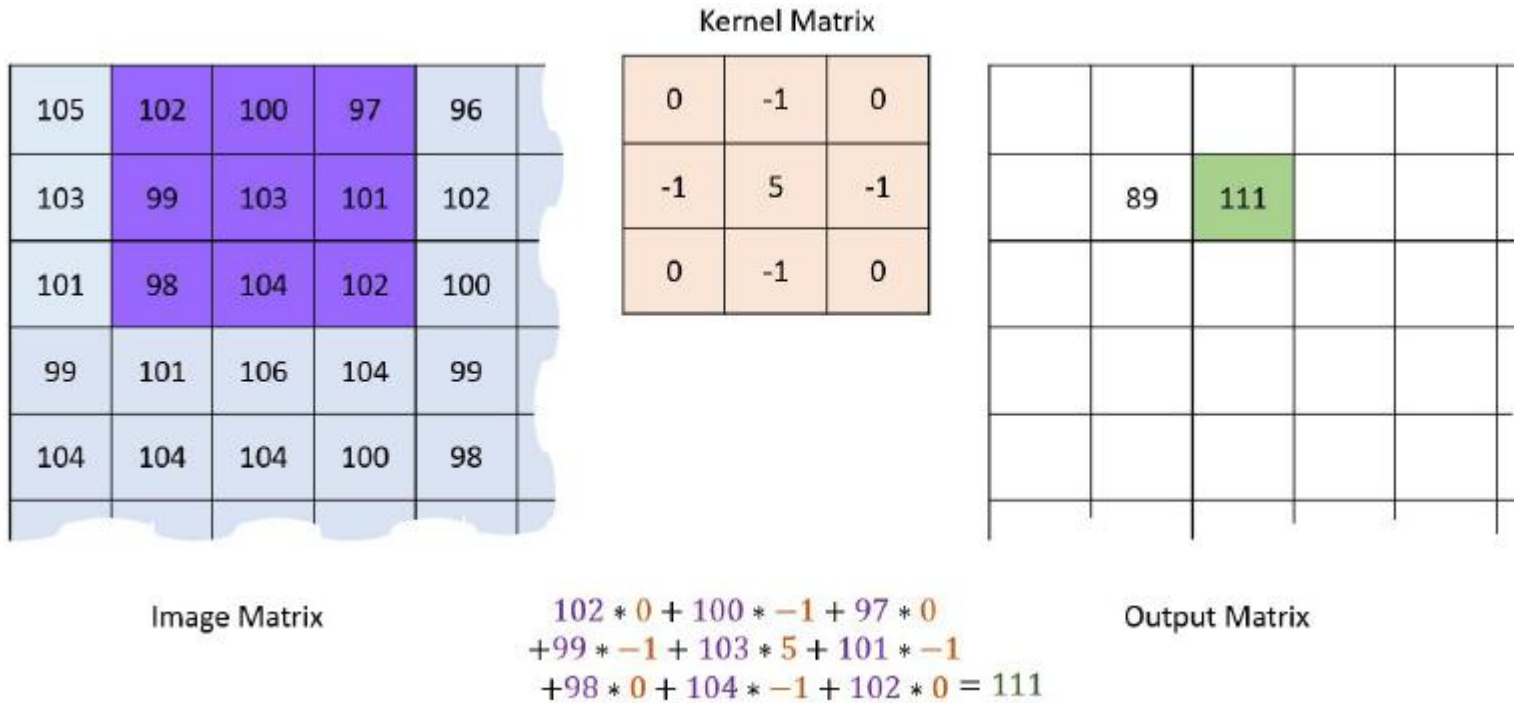
Source: Machinelearningguru.com/computer_vision/basics/convolution/image_convolution_1.html

Image convolution operation



Source: [Machinelearningguru.com/computer_vision/basics/convolution/image_convolution_1.html](https://machinelearningguru.com/computer_vision/basics/convolution/image_convolution_1.html)

Image convolution operation



Source: [Machinelearningguru.com/computer_vision/basics/convolution/image_convolution_1.html](https://machinelearningguru.com/computer_vision/basics/convolution/image_convolution_1.html)

Convolution

Input matrix

X_1	X_2	X_3	X_4
X_5	X_6	X_7	X_8
X_9	X_{10}	X_{11}	X_{12}
X_{13}	X_{14}	X_{15}	X_{16}

Size: 4 x 4

convolution
*

filter

W_1	W_2	W_3
W_4	W_5	W_6
W_7	W_8	W_9

Size: 3 x 3

Output matrix ?

Convolution

Input matrix

X_1	X_2	X_3	X_4
X_5	X_6	X_7	X_8
X_9	X_{10}	X_{11}	X_{12}
X_{13}	X_{14}	X_{15}	X_{16}

Size: 4 x 4

filter

W_1	W_2	W_3
W_4	W_5	W_6
W_7	W_8	W_9

Size: 3 x 3

convolution
*

Output matrix

Z_1	Z_2
Z_3	Z_4

Size: 2 x 2

=

Convolution

Input matrix

X_1	X_2	X_3	X_4
X_5	X_6	X_7	X_8
X_9	X_{10}	X_{11}	X_{12}
X_{13}	X_{14}	X_{15}	X_{16}

Size: 4 x 4

filter

W_1	W_2
W_3	W_4

Size: 2 x 2

convolution
*

Output matrix

Z_1	Z_2	Z_3
Z_4	Z_5	Z_6
Z_7	Z_8	Z_9

=
Size: 3 x 3

Convolution operation before activation

X_1	X_2	X_3	X_4
X_5	X_6	X_7	X_8
X_9	X_{10}	X_{11}	X_{12}
X_{13}	X_{14}	X_{15}	X_{16}

 *

W_1	W_2	W_3
W_4	W_5	W_6
W_7	W_8	W_9

 =

Z_1	Z_2
Z_3	Z_4

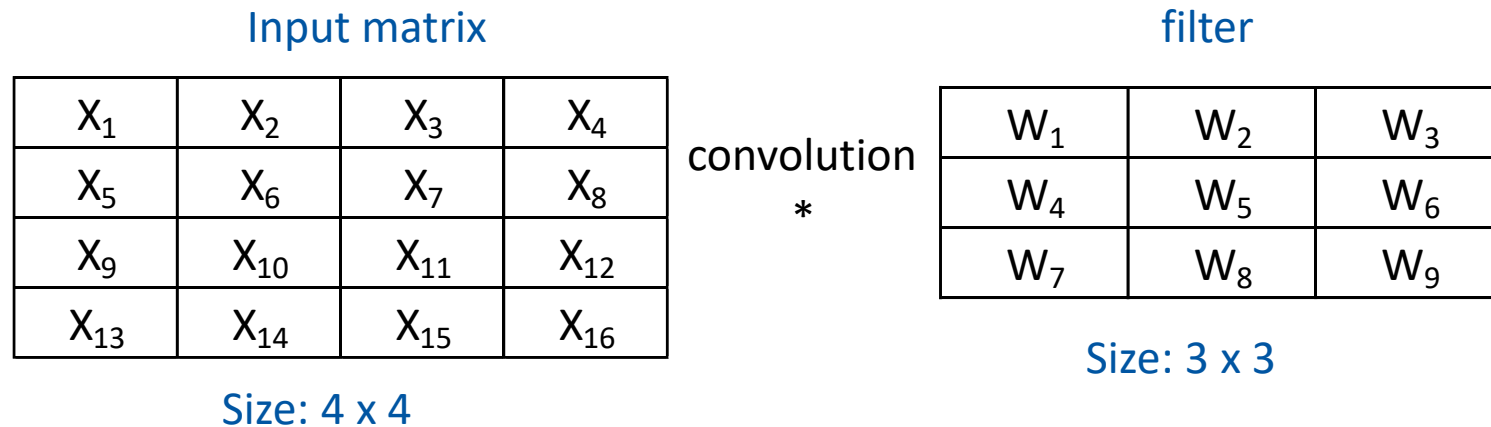
$$\begin{aligned} Z_1 = & X_1 \times W_1 + X_2 \times W_2 + X_3 \times W_3 \\ & + X_5 \times W_4 + X_6 \times W_5 + X_7 \times W_6 \\ & + X_9 \times W_7 + X_{10} \times W_8 + X_{11} \times W_9 \end{aligned}$$

$$\begin{aligned} Z_2 = & X_2 \times W_1 + X_3 \times W_2 + X_4 \times W_3 \\ & + X_6 \times W_4 + X_7 \times W_5 + X_8 \times W_6 \\ & + X_{10} \times W_7 + X_{11} \times W_8 + X_{12} \times W_9 \end{aligned}$$

$$\begin{aligned} Z_3 = & X_5 \times W_1 + X_6 \times W_2 + X_7 \times W_3 \\ & + X_9 \times W_4 + X_{10} \times W_5 + X_{11} \times W_6 \\ & + X_{13} \times W_7 + X_{14} \times W_8 + X_{15} \times W_9 \end{aligned}$$

$$\begin{aligned} Z_4 = & X_6 \times W_1 + X_7 \times W_2 + X_8 \times W_3 \\ & + X_{10} \times W_4 + X_{11} \times W_5 + X_{12} \times W_6 \\ & + X_{14} \times W_7 + X_{15} \times W_8 + X_{16} \times W_9 \end{aligned}$$

Convolution operation in activation

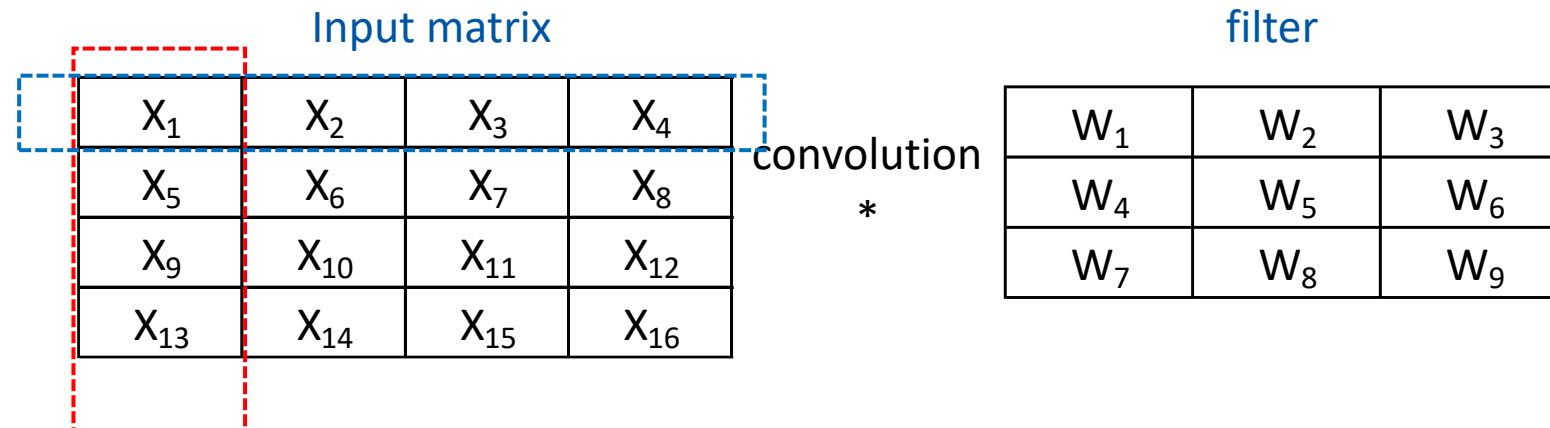


g : activation function
 b : bias

$$= \begin{bmatrix} g(Z_1 + b) & g(Z_2 + b) \\ g(Z_3 + b) & g(Z_4 + b) \end{bmatrix} = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}$$

Size: 2 x 2

Edge pixels



What about information from edge pixels?

Padding

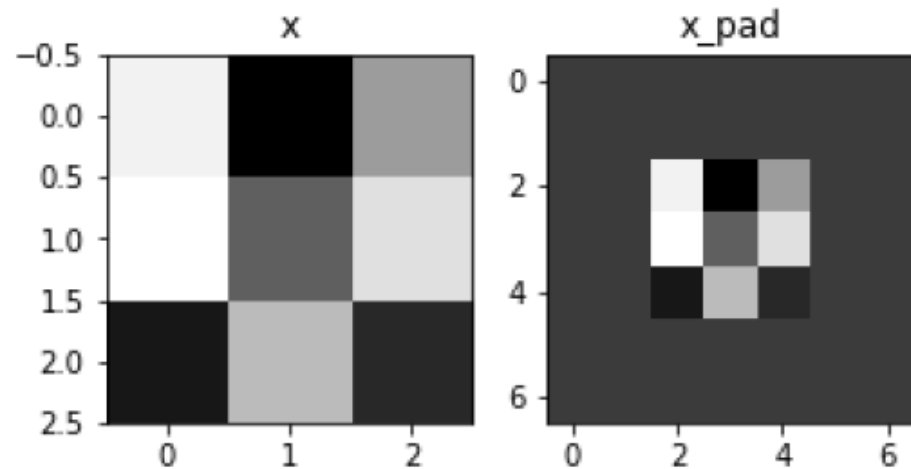
To detect the information of edge or border

To prevent shrinking(compression) of size of next layer

Zero pad

Input: (m, n_H, n_W, n_C)

Output: (m, n_H + 2*pad, n_W + 2*pad, n_C)



Source: Deep Learning, Andrew Ng

Zero pad

np.pad

```
In [4]: a
```

```
Out[4]: array([[0, 1, 2],  
               [3, 4, 5]])
```

```
In [7]: np.pad(a, (1,1), 'constant', constant_values=(0))
```

```
Out[7]: array([[0, 0, 0, 0, 0],  
               [0, 0, 1, 2, 0],  
               [0, 3, 4, 5, 0],  
               [0, 0, 0, 0, 0]])
```

```
In [8]: np.pad(a, (2,2), 'constant', constant_values=(0))
```

```
Out[8]: array([[0, 0, 0, 0, 0, 0, 0],  
               [0, 0, 0, 0, 0, 0, 0],  
               [0, 0, 0, 1, 2, 0, 0],  
               [0, 0, 3, 4, 5, 0, 0],  
               [0, 0, 0, 0, 0, 0, 0],  
               [0, 0, 0, 0, 0, 0, 0]])
```

Padding

0	0	0	0	0	0
0	x_1	x_2	x_3	x_4	0
0	x_5	x_6	x_7	x_8	0
0	x_9	x_{10}	x_{11}	x_{12}	0
0	x_{13}	x_{14}	x_{15}	x_{16}	0
0	0	0	0	0	0

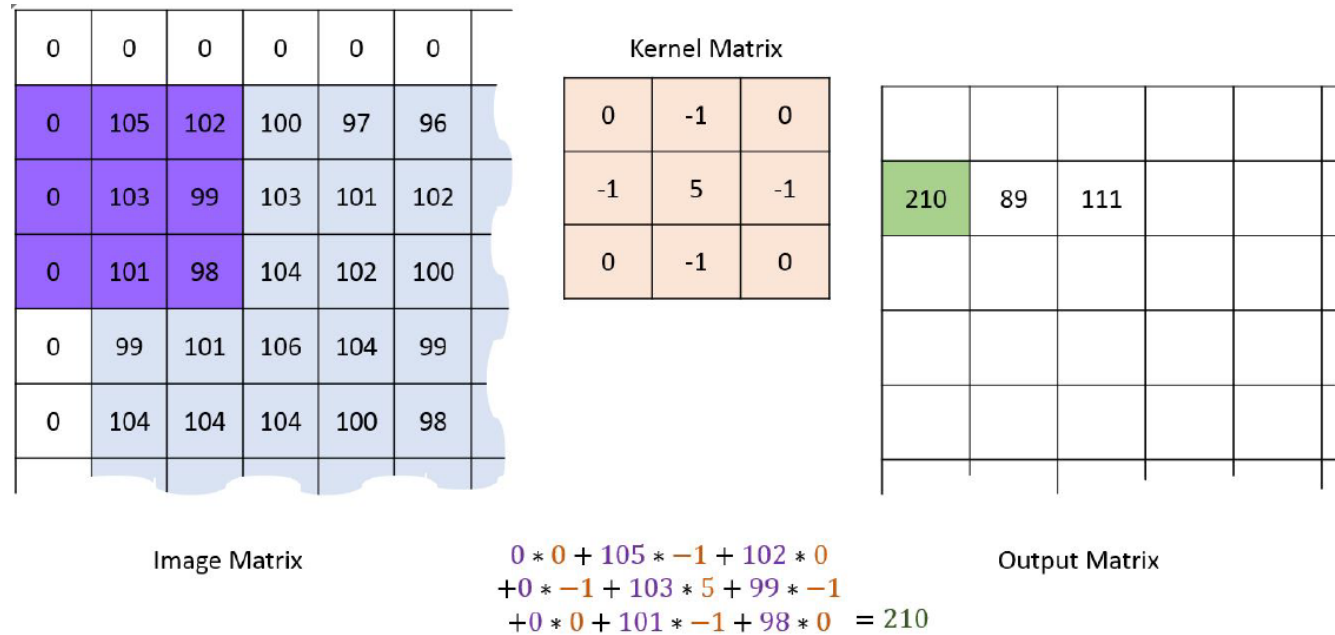
*

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

=

z_1	z_2	z_3	z_4
z_5	z_6	z_7	z_8
z_9	z_{10}	z_{11}	z_{12}
z_{13}	z_{14}	z_{15}	z_{16}

Convolution operation on borders



Source: [Machinelearninguru.com/computer_vision/basics/convolution/image_convolution_1.html](https://machinelearninguru.com/computer_vision/basics/convolution/image_convolution_1.html)

Convolution operation on borders

0	0	0	0	0	0
0	105	102	100	97	96
0	103	99	103	101	102
0	101	98	104	102	100
0	99	101	106	104	99
0	104	104	104	100	98

Image Matrix

0	-1	0
-1	5	-1
0	-1	0

Kernel Matrix

320				
210	89	111		

Output Matrix

$$\begin{aligned} &0 * 0 + 0 * -1 + 0 * 0 \\ &+ 0 * -1 + 105 * 5 + 102 * -1 \\ &+ 0 * 0 + 103 * -1 + 99 * 0 = 320 \end{aligned}$$

Source: [Machinelearningguru.com/computer_vision/basics/convolution/image_convolution_1.html](https://machinelearningguru.com/computer_vision/basics/convolution/image_convolution_1.html)

Padding: 1 zero padding

0	0	0	0	0	0
0	X_1	X_2	X_3	X_4	0
0	X_5	X_6	X_7	X_8	0
0	X_9	X_{10}	X_{11}	X_{12}	0
0	X_{13}	X_{14}	X_{15}	X_{16}	0
0	0	0	0	0	0

*

W_1	W_2	W_3
W_4	W_5	W_6
W_7	W_8	W_9

=

Z_1	Z_2	Z_3	Z_4
Z_5	Z_6	Z_7	Z_8
Z_9	Z_{10}	Z_{11}	Z_{12}
Z_{13}	Z_{14}	Z_{15}	Z_{16}

$$\begin{aligned} Z_1 &= 0 \times W_1 + 0 \times W_2 + 0 \times W_3 \\ &\quad + 0 \times W_4 + X_1 \times W_5 + X_2 \times W_6 \\ &\quad + 0 \times W_7 + X_5 \times W_8 + X_6 \times W_9 \end{aligned}$$

$$\begin{aligned} Z_4 &= 0 \times W_1 + 0 \times W_2 + 0 \times W_3 \\ &\quad + X_3 \times W_4 + X_4 \times W_5 + 0 \times W_6 \\ &\quad + X_7 \times W_7 + X_8 \times W_8 + 0 \times W_9 \end{aligned}$$

Padding: 2 zero padding

0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	X_1	X_2	X_3	X_4	0	0
0	0	X_5	X_6	X_7	X_8	0	0
0	0	X_9	X_{10}	X_{11}	X_{12}	0	0
0	0	X_{13}	X_{14}	X_{15}	X_{16}	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

*

W_1	W_2	W_3
W_4	W_5	W_6
W_7	W_8	W_9

Valid vs. Same convolution

- Valid convolution: without padding

$$(n \times n) * (f \times f) \rightarrow (n-f+1) \times (n-f+1)$$

$$(6 \times 6) * (3 \times 3) \rightarrow (4 \times 4)$$

- Same convolution: with zero padding, output size is the same as the input size

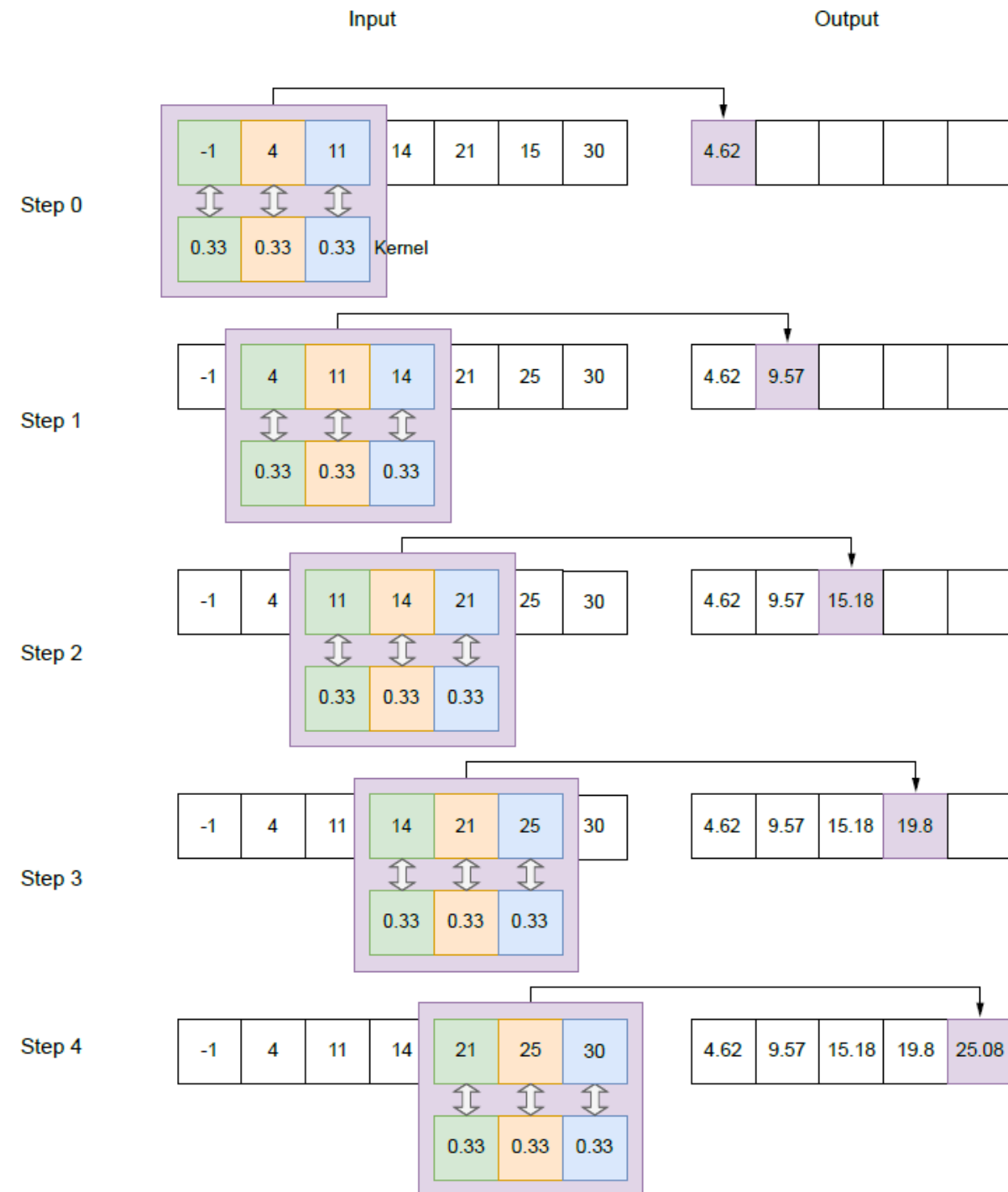
$$(n \times n) * (f \times f) \rightarrow (n+2p-f+1) \times (n+2p-f+1)$$

$$(6 \times 6) * (3 \times 3) \rightarrow (6 \times 6)$$

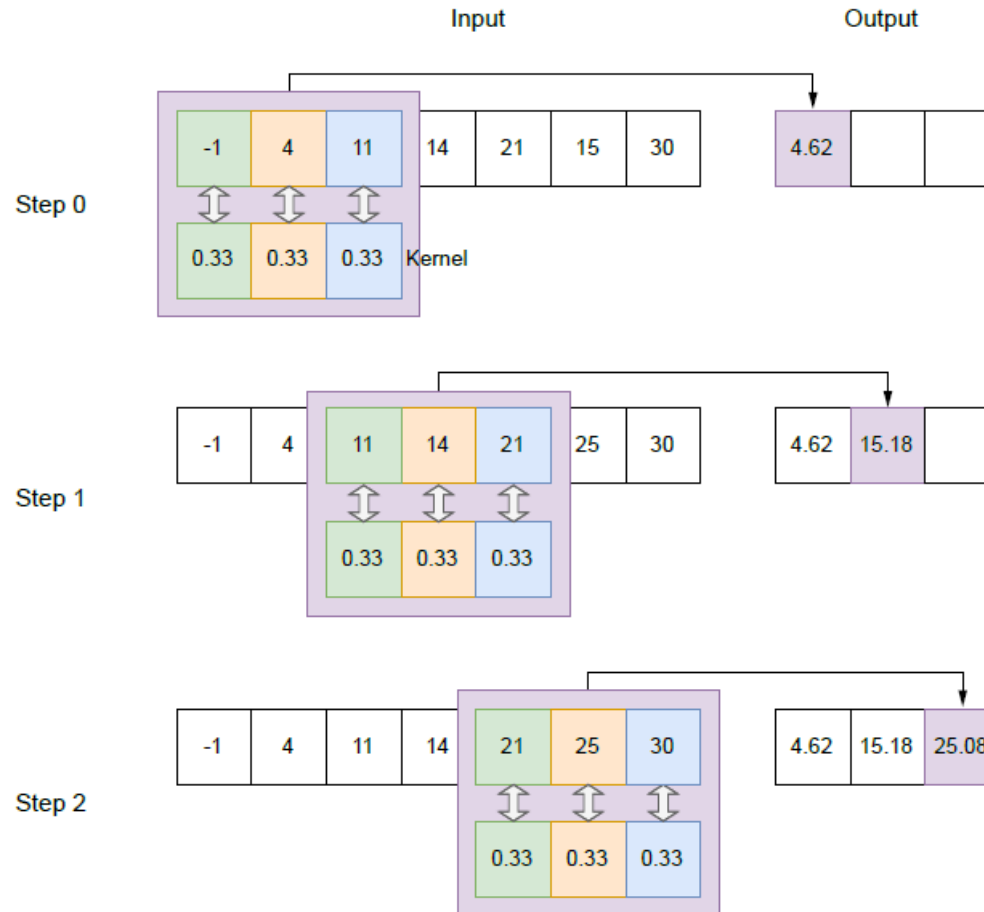
Stride

To compress the size of next layer

1D convolution with stride 1



1D convolution with stride 2



Convolution operation with stride 1

0	0	0	0	0	0
0	105	102	100	97	96
0	103	99	103	101	102
0	101	98	104	102	100
0	99	101	106	104	99
0	104	104	104	100	98

Kernel Matrix		
0	-1	0
-1	5	-1
0	-1	0

320	206	198	188	
210	89	111		

Image Matrix

$$\begin{aligned} &102 * 0 + 100 * -1 + 97 * 0 \\ &+ 99 * -1 + 103 * 5 + 101 * -1 \\ &+ 98 * 0 + 104 * -1 + 102 * 0 = 111 \end{aligned}$$

Output Matrix

Source: [Machinelearningguru.com/computer_vision/basics/convolution/convolution_layer.html](https://machinelearningguru.com/computer_vision/basics/convolution/convolution_layer.html)

Convolution operation with stride 2

0	0	0	0	0	0	0
0	105	102	100	97	96	0
0	103	99	103	101	102	10
0	101	98	104	102	100	1
0	99	101	106	104	99	0
0	104	104	104	100	98	1

Kernel Matrix		
0	-1	0
-1	5	-1
0	-1	0

320	198	182		

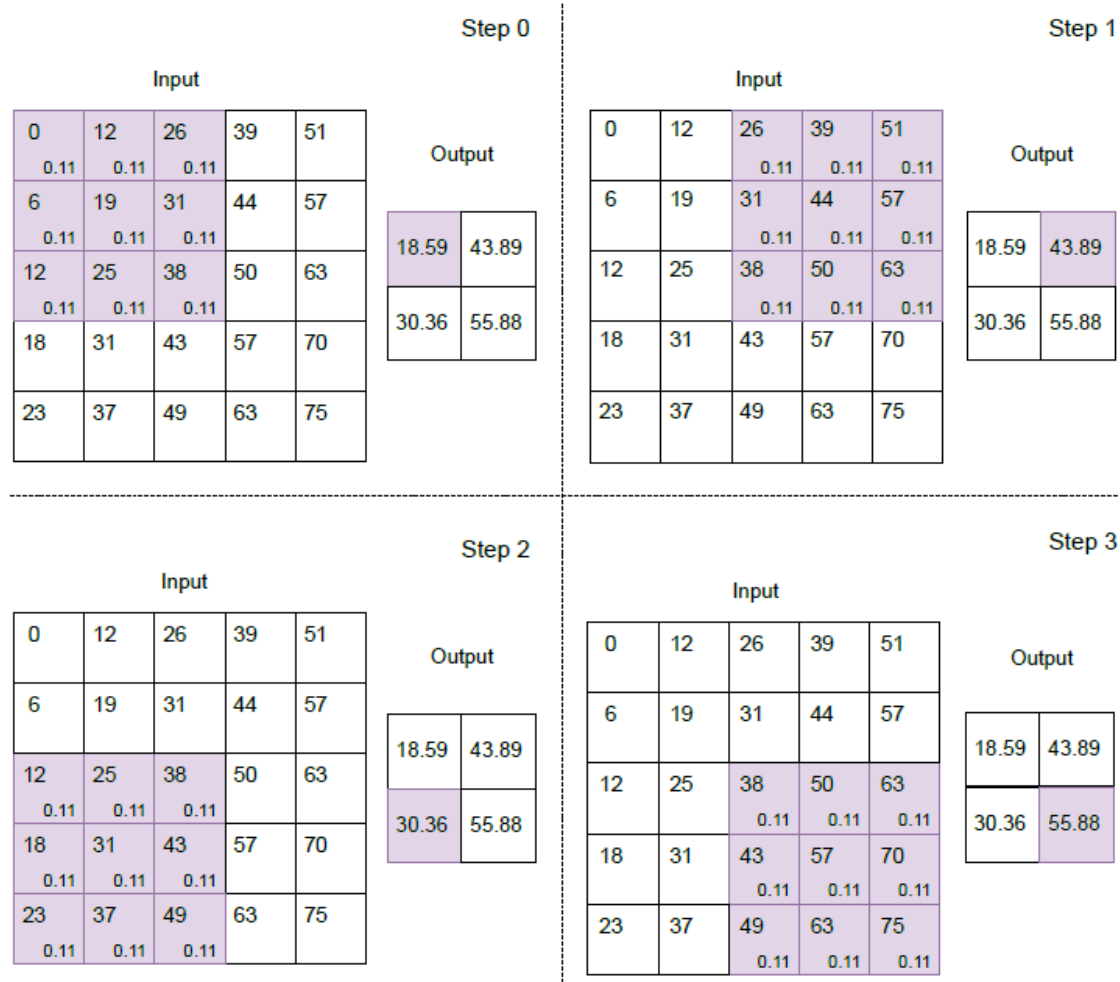
Image Matrix

$$\begin{aligned} &0 * 0 + 0 * -1 + 0 * 0 \\ &+ 97 * -1 + 96 * 5 + 99 * -1 \\ &+ 101 * 0 + 102 * -1 + 101 * 0 = 182 \end{aligned}$$

Output Matrix

Source: [Machinelearningguru.com/computer_vision/basics/convolution/convolution_layer.html](https://machinelearningguru.com/computer_vision/basics/convolution/convolution_layer.html)

2D convolution with stride 2, valid padding



Source: Math and Architectures of Deep Learning, Krishnedu Chauhury

Convolutions on RGB images

0	0	0	0	0	0	...
0	156	155	156	158	158	...
0	153	154	157	159	159	...
0	149	151	153	158	159	...
0	146	146	149	153	158	...
0	145	143	143	148	158	...
...	...	...	...	...	...	...

Input Channel #1 (Red)

0	0	0	0	0	0	...
0	167	166	167	169	169	...
0	164	165	168	170	170	...
0	160	162	166	169	170	...
0	156	156	159	163	168	...
0	155	153	153	158	168	...
...	...	...	...	...	...	...

Input Channel #2 (Green)

0	0	0	0	0	0	...
0	163	162	163	165	165	...
0	160	161	164	166	166	...
0	156	158	162	165	166	...
0	155	155	158	162	167	...
0	154	152	152	157	167	...
...	...	...	...	...	...	...

Input Channel #3 (Blue)

W

1	0	0
0	-1	0
0	0	1

Size: 3 x 3

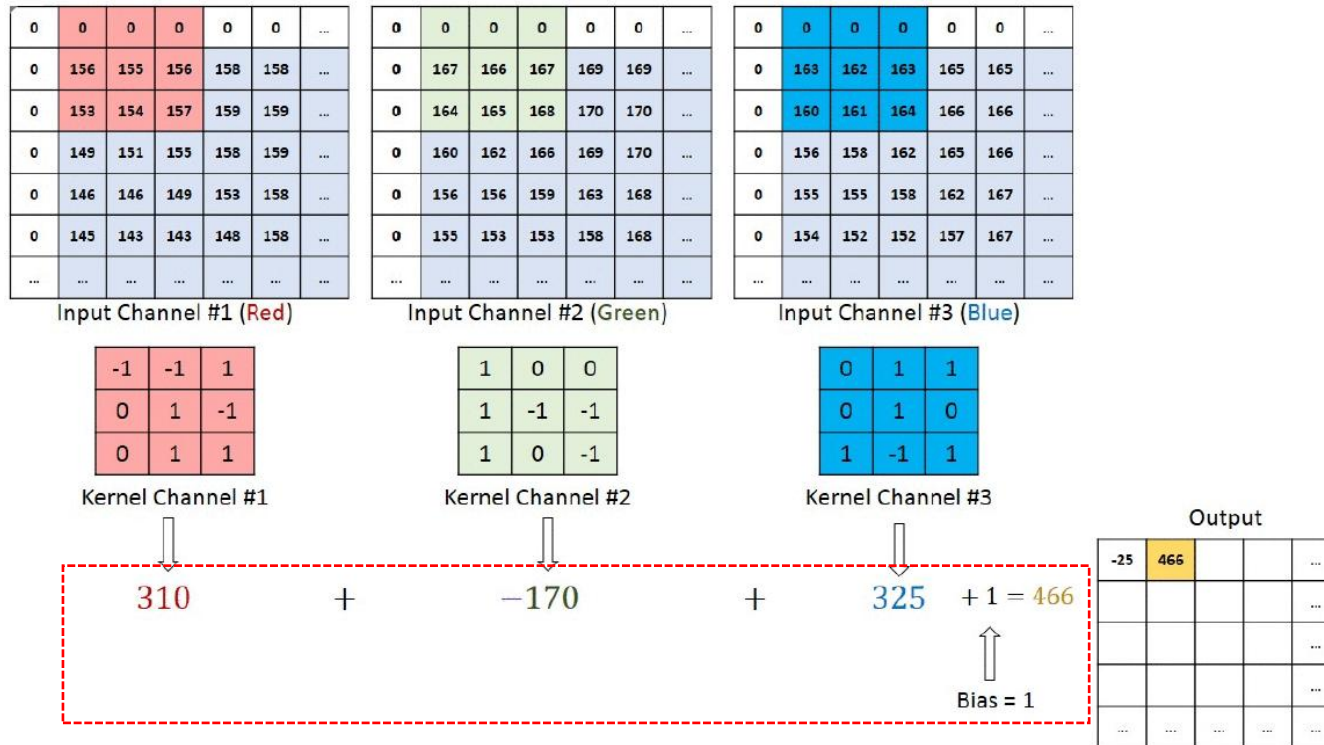
$$Z = W*a(\text{Red})+W*a(\text{Green})+W*a(\text{Blue})+b$$

$$(0-155+157)+(0-166+168)+(0-162+164)+5$$

$$= 1$$

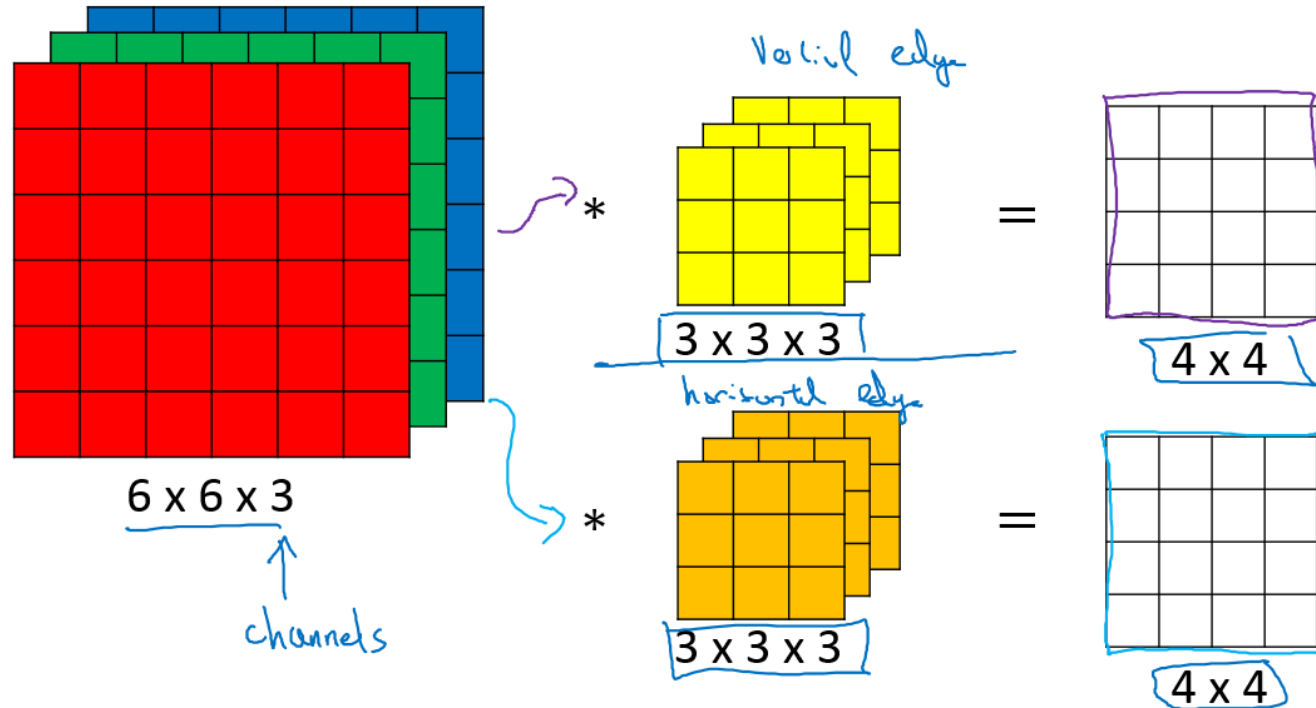
Source: [Machinelearningguru.com/computer_vision/basics/convolution/convolution_layer.html](https://machinelearningguru.com/computer_vision/basics/convolution/convolution_layer.html)

Convolutions on RGB images



Source: [Machinelearningguru.com/computer_vision/basics/convolution/convolution_layer.html](https://machinelearningguru.com/computer_vision/basics/convolution/convolution_layer.html)

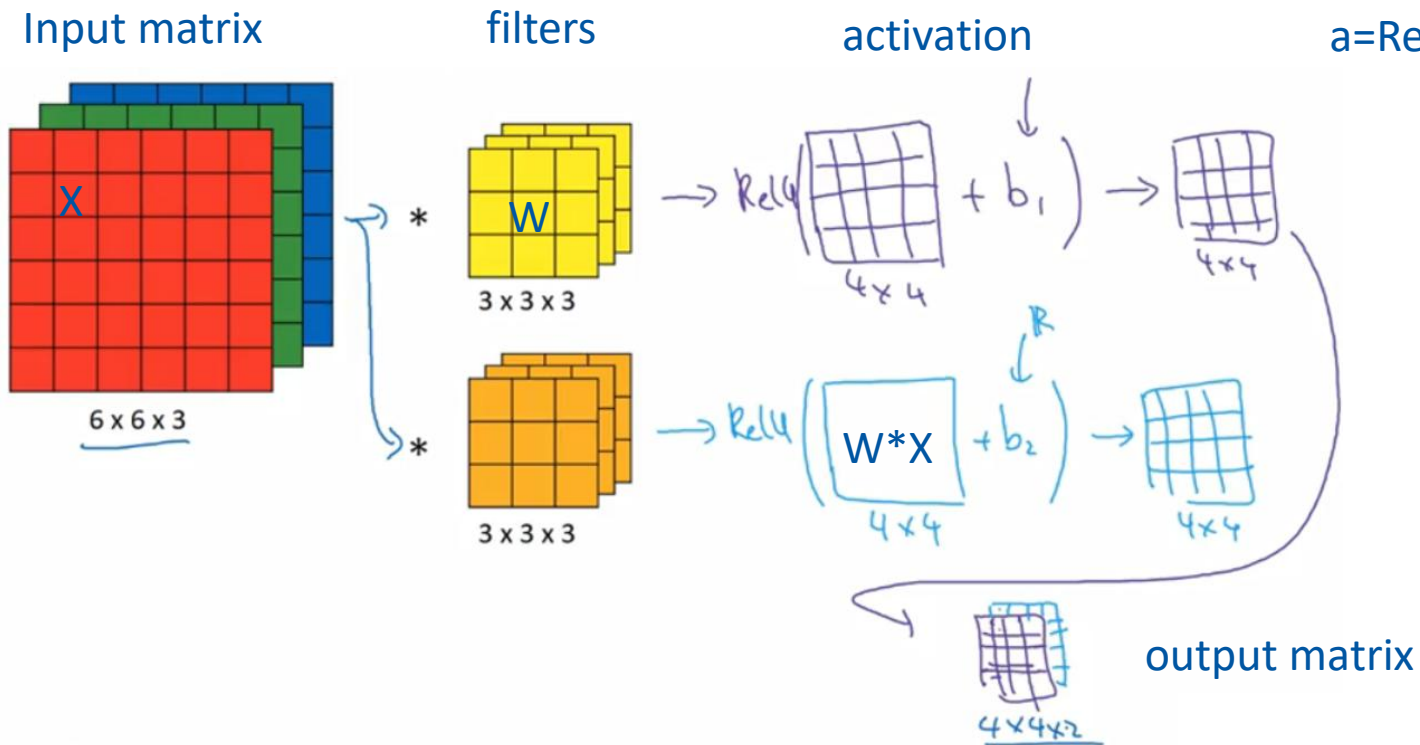
Multiple filters (channels)



Source: deep learning, Andrew Ng

Multiple filters (channels)

$$Z = W * X + b$$
$$a = \text{ReLU}(Z)$$



Source: deep learning, Andrew Ng

Output size after convolution

- $n \times n$ image
- $k \times k$ filter
- padding p
- stride s

Output size:

$$\left\lceil \frac{n + 2p - k}{s} + 1 \right\rceil \times \left\lceil \frac{n + 2p - k}{s} + 1 \right\rceil$$

k is usually odd number in computer vision.

Number of parameters in a convolution layer

If *layer l* is a convolution layer:

$f^{[l]}$ = filter size

$p^{[l]}$ = padding

$s^{[l]}$ = stride

Input size:

$$n_H^{[l-1]} \times n_W^{[l-1]} \times n_c^{[l-1]}$$

output size:

$$n_H^{[l]} \times n_W^{[l]} \times n_c^{[l]}$$

$$n_H^{[l]} = \frac{n_H^{[l-1]} + 2p^{[l]} - f^{[l]}}{s^{[l]}} + 1$$

$$n_W^{[l]} = \frac{n_W^{[l-1]} + 2p^{[l]} - f^{[l]}}{s^{[l]}} + 1$$

Weights: $f^{[l]} \times f^{[l]} \times n_c^{[l-1]} \times n_c^{[l]}$

bias: $1 \times 1 \times 1 \times n_c^{[l]}$

Pooling

No parameters to learn

- To subsample (shrink) the input image in order to reduce the computational load, the memory usage, the number of parameters (reduce overfitting)

Pooling

2	3	1	9
4	7	3	5
8	2	2	2
1	3	4	5

Hyperparameters

$f=2$
 $s=2$

Max:

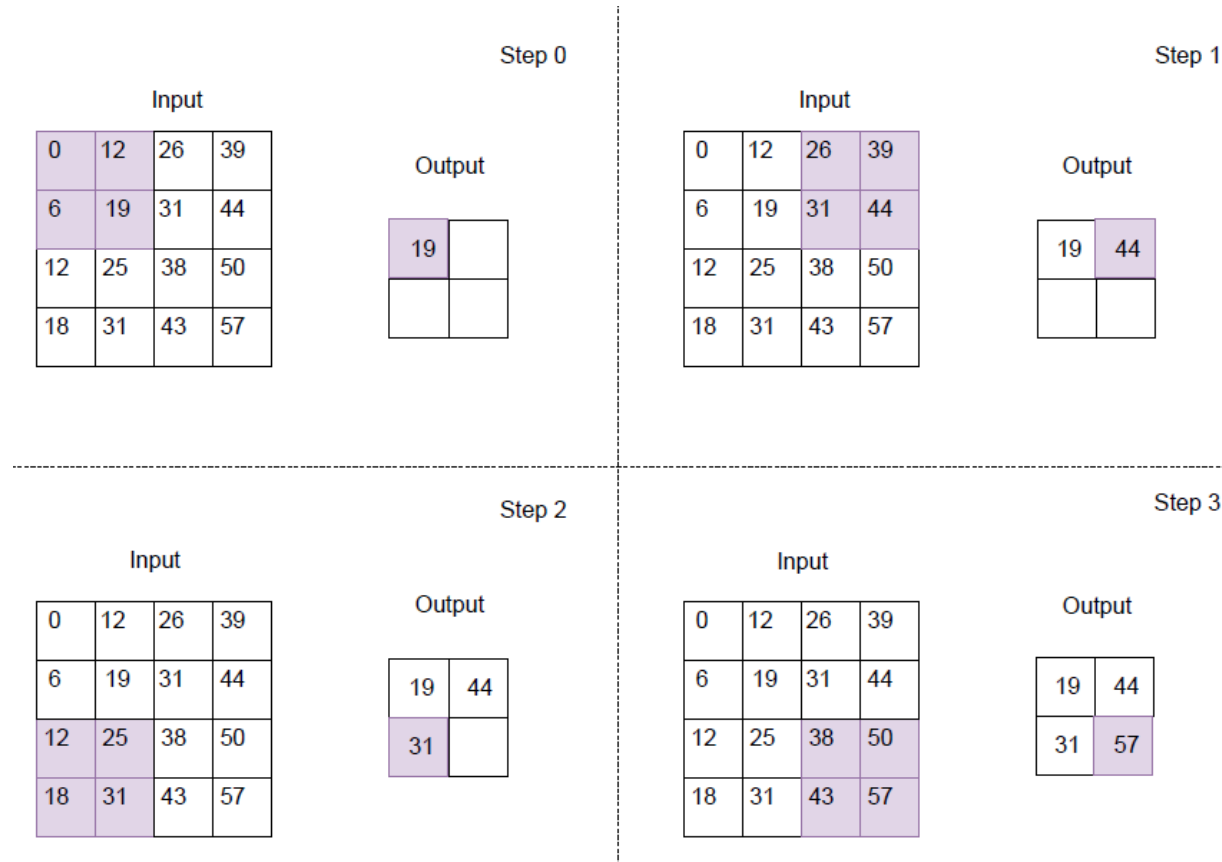
7	9
8	5

Average:

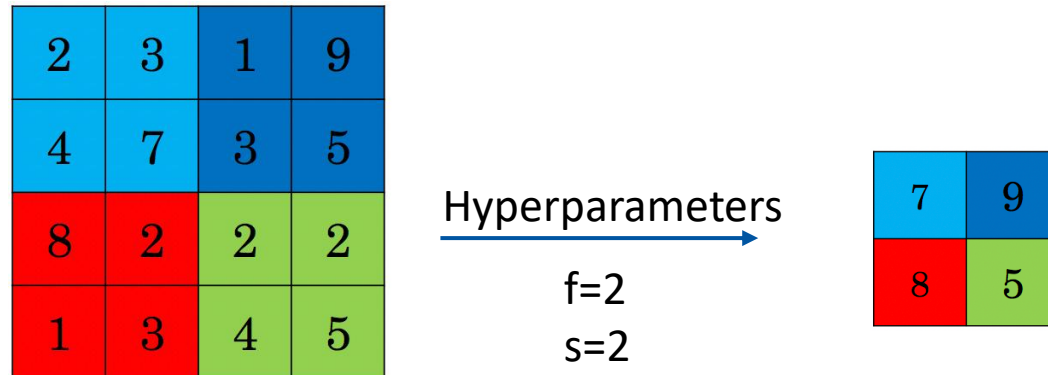
4	4.5
3.25	3.25

Source: Deep Learning, Andrew Ng

Max Pooling with 2x2 kernel and stride 2

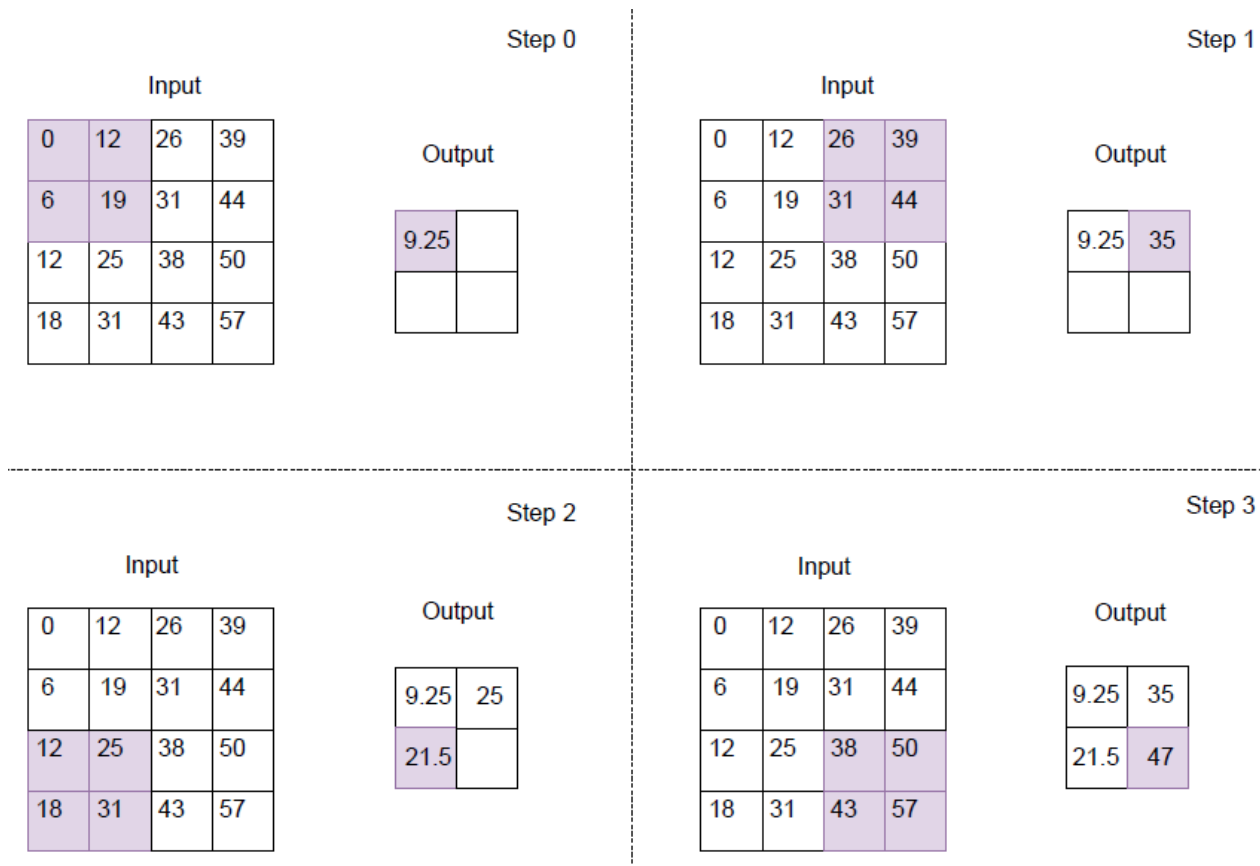


Max Pooling

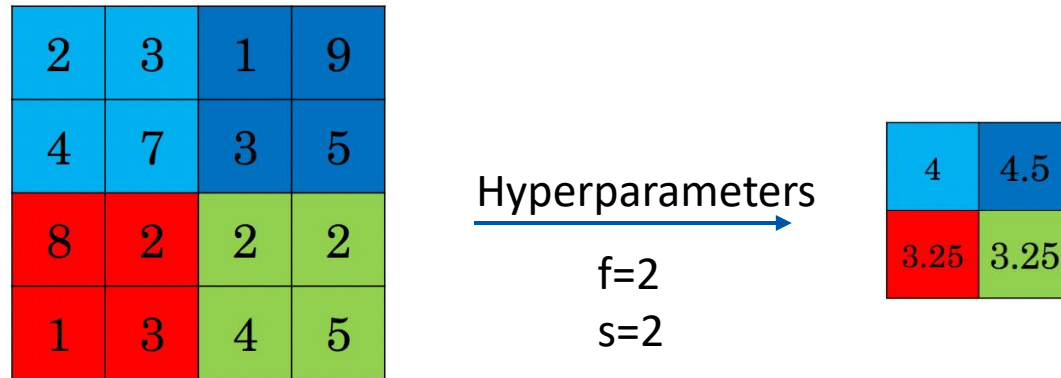


Source: Deep Learning, Andrew Ng

Average Pooling with 2x2 kernel and stride 2



Average Pooling



Source: Deep Learning, Andrew Ng

Size of Pooling

- Hyperparameters
 - f: filter size
 - s : stride
 - p : padding (usually p=0)
- Max or average pooling
- No parameters to learn

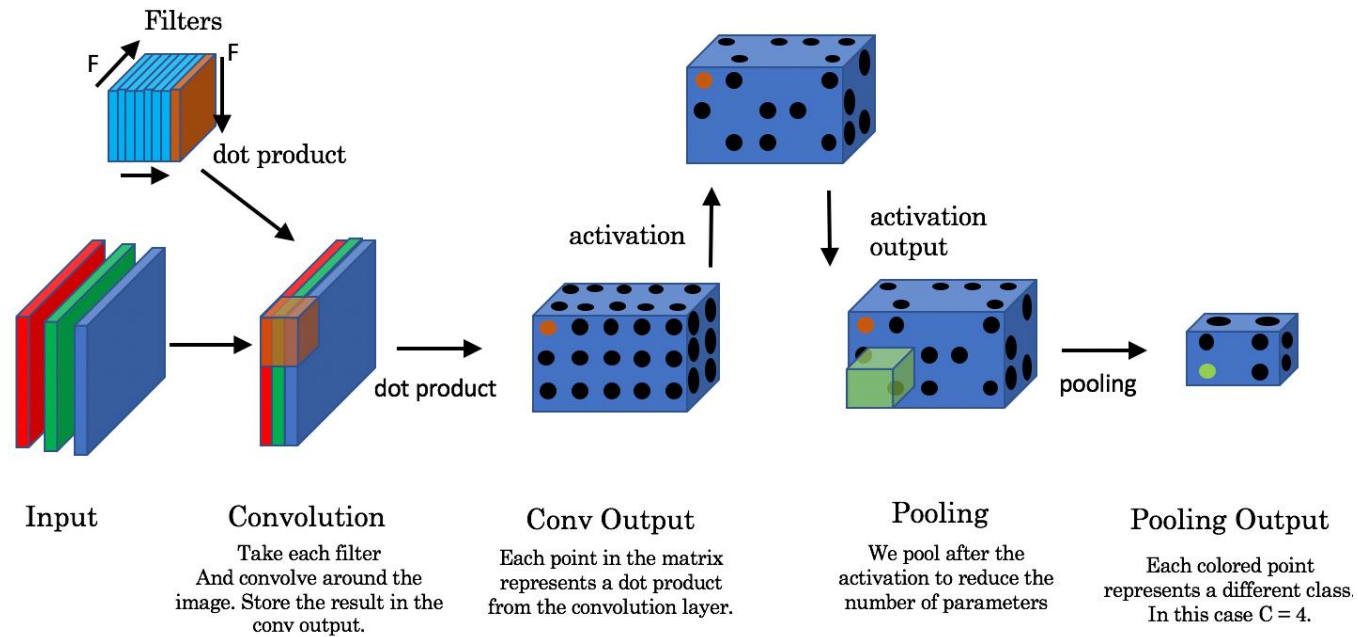
Input size: $n_H^{[l-1]} \times n_W^{[l-1]} \times n_c^{[l-1]}$

output size: $n_H^{[l]} \times n_W^{[l]} \times n_c^{[l]}$

$$n_H^{[l]} = \frac{n_H^{[l-1]} - f^{[l]}}{s^{[l]}} + 1$$

$$n_W^{[l]} = \frac{n_W^{[l-1]} - f^{[l]}}{s^{[l]}} + 1$$

Convolution and Pooling



Source: Deep Learning, Andrew Ng